Workshop Proposal for ECML-PKDD 2026

5th Workshop on Machine Learning for Irregular Time Series: Advances in Generative Models, Global Models and Self-Supervised Learning [**ML4ITS 2026**]

Brief Description

Machine Learning for Irregular Time Series (ML4ITS2026) focuses on the development of models and learning paradigms capable of handling the complexity of real-world time series data. Unlike classical benchmark datasets with regular sampling and abundant labels, real operational data is often characterized by **irregular sampling, missing values, multi-resolution signals, noise, non-stationarity, heterogeneity, and scarce supervision**.

Irregular time series arise naturally in mission-critical domains including finance, healthcare, environmental monitoring, industrial systems, space operations, and IoT. In these settings, reliable modeling is essential for forecasting, anomaly detection, decision support, and autonomous control. However, traditional statistical approaches and standard deep learning pipelines often fail under irregularity, limited data regimes, and domain shifts. ML4ITS2026 addresses these challenges by promoting research at the intersection of:

- Data-efficient deep learning
- Generative modeling for time series
- Self-supervised and representation learning
- Foundation models for temporal data
- Trustworthy and responsible AI (uncertainty and explainability)

A central premise of the workshop is that **irregularity is the norm, not the exception** in real-world time series analysis. We define time series as *irregular* if they exhibit one or more of the following characteristics:

- **Short** – limited historical data and small sample sizes.
- **Multiresolution** – varying sampling frequencies across variables.
- **Noisy and Incomplete** – missing data, corruption, or measurement errors.
- **Heterogeneous** – signals originating from diverse physical systems with distinct statistical structures.
- **Scarcely Labeled or Unlabeled** – limited or absent annotations.

In practice, two challenging scenarios dominate. The first is that **data is limited and fragmented** and the second is that **large volumes of data exist, but labels are scarce or costly**. These realities motivate AI systems that can *do more with less* — leveraging self-supervision, transfer learning, generative modeling, and scalable pretraining strategies while respecting privacy, sustainability, and robustness constraints.

Building on the **success of previous ML4ITS editions at ECML-PKDD (2021, 2023, 2024, 2025)**, the 2026 workshop aims to further consolidate a community focused specifically on irregular, heterogeneous, and data-scarce temporal regimes that are often underrepresented in

the main conference track. We invite contributions addressing irregularity aspects (a)-(e) and encourage submissions on:

- Generative models (e.g., GANs, diffusion models, masked modeling) for Synthetic Data generation.
- Explainable AI techniques tailored to deep time series models.
- Uncertainty quantification in deep time series models.
- Methods for Data Imputation and Denoising.
- Transfer Learning for Time Series forecasting and classification, including FNN, CNN, and Recurrent NN.
- Transformers architectures and Attention mechanisms for Time Series analysis.
- Graph Neural Networks for Anomaly Detection and Failure Prediction.
- Deep Neural Networks (e.g., FNN, CNN, Recurrent NN, LSTMs) for Time Series modeling and forecasting.
- Unsupervised and Self-Supervised Learning for various Time Series tasks.
- Few-Shot Learning and Time Series Classification in low-data scenarios.
- Physical-informed Deep Neural Networks for Time Series Forecasting.
- (Deep) Reservoir Computing and Spiking Neural Networks for Time Series and Structured data analysis.
- Representation Learning for Time Series.
- Time series with a large number of channels, or changing number of channels.

This workshop will concentrate on four specific areas: A) generative models for time series, B) self-supervised learning for time series C) responsible AI for time series, and D) Foundation models for time series. This workshop focuses on these areas for the following reasons:
A) **Generative models for time series**: Generative models enable the synthesis of new time series data, which can be employed to augment existing datasets, perform data imputation, or even simulate various scenarios. Developing more advanced generative models specific to time series can lead to a better understanding of underlying temporal patterns, which, in turn, can improve forecasting, anomaly detection, and decision-making in fields like finance, healthcare, and environmental science.
B) **Self-supervised learning for time series**: In real-world situations, obtaining labeled data can be challenging, time-consuming, and expensive. Self-supervised learning techniques can leverage the vast amounts of unlabeled data available to improve model performance without relying on manual annotation. Researchers can develop more efficient and scalable models by focusing on self-supervised learning methods for time series, ultimately leading to better insights and predictions even with limited labeled data.
C) **Responsible AI for time series:** In response to the mission-critical nature of many time series applications, it is natural to consider responsible aspects of these models. In particular, Explainable AI and uncertainty quantification are themes of key importance both for decision makers informed by models (such as central banks, health providers, etc.) and automated systems (industrial manufacturing, quantitative trading, etc.)
D) **Foundation Models for Time Series:** The emergence of foundation models, pre-trained on vast, diverse data, marks a significant shift in machine learning. These models excel in

generalization, outperforming traditional methods confined to smaller datasets. This approach can also be transformative in time series analysis, where extensive training on global, multimodal data enhances accuracy and applicability across various domains.

Special Session - “Time series analysis for space applications”

Machine learning is a key enabler for autonomous space missions and scalable space operations. Space operations increasingly rely on AI, and machine learning specifically, for mission-critical tasks, such as anomaly detection, FDIR (Fault Detection, Isolation and Restoration), satellite health monitoring, mission planning, space weather forecasting, or flight dynamics. All these topics are part of the larger Artificial Intelligence for Automation (A2I) Roadmap initiative of the European Space Agency’s (ESA) European Space Operations Center (ESOC) that aims to bring AI to automate ground segment operations (https://esoc.esa.int/a2i-roadmap-0). For example, monitoring spacecraft telemetry (collection of continuous measurements and onboard instrument readings) is crucial for mission safety, planning, and autonomy. Spacecraft telemetry is also a perfect example of complex, irregular, multivariate time series. It features irregular sampling, data gaps, outliers (due to measurement errors), complex interdependencies, large data volume, and dynamically evolving behaviours based on spacecraft status, orbit, and attitude, to name a few.
To support development and evaluation of algorithms for spacecraft telemetry analysis, ESA, in collaboration with Airbus Defence and Space and KP Labs, released two curated real-life datasets: ESA-ADB (7 GB of raw data, several years of telemetry from 3 missions, with carefully annotated anomalies) and OPSSAT-AD (a smaller set of annotated fragments from a single mission) - see Tables 1 and 2 for references.

ESA Benchmark for Anomaly Detection in Satellite Telemetry (ESA-ADB)	
Article	https://arxiv.org/abs/2406.17826
Dataset	https://zenodo.org/records/12528696
Benchmarking Code	https://github.com/kplabs-pl/ESA-ADB

Table 1 - Details of the ESA-ADB benchmark

OPS-SAT Benchmark for Anomaly Detection in Satellite Telemetry (OPSSAT-AD)	
Article	https://doi.org/10.1038/s41597-025-05035-3
Dataset	https://zenodo.org/records/12588359
Benchmarking Code	https://github.com/kplabs-pl/OPS-SAT-AD

Table 2 - Details of the OPSSAT-AD benchmark

The larger ESA-ADB dataset was a foundation for the successful Kaggle Challenge for anomaly detection in satellite telemetry, summarized in the previous edition of ML4ITS (https://ai4gs.space-codev.org/esa-adb/competitions/). Its main goal is to encourage researchers from academia, research institutes, and the space industry to propose new effective algorithms to improve anomaly detection in practical settings of space operations.

The dedicated session for space applications will cover the following topics:

- Time series analysis for satellite telemetry
- Anomaly detection in multivariate time series
- Time series forecasting for satellite health monitoring
- Time series data compression to save communication bandwidth
- Foundation models for satellite telemetry data
- Time series analysis optimized for spacecraft on-board applications
- Explainability and trustworthiness of time series analysis in space operations
- Benchmarking of time series analysis methods
- Predictive maintenance in spacecraft applications

The session's relevance goes beyond the space community as the discussed topics have far-reaching implications across various sectors sharing similar challenges to the space domain, such as finance, healthcare, robotics, IoT, automotive, communication, aerospace, and defence.

Contact Information / Organizers

Massimiliano Ruocco (NTNU/Sintef, Norway) is a Senior Researcher at Sintef Digital and Associate Professor of Machine Learning at the Norwegian University of Science and Technology (NTNU) in the Data and Artificial Intelligence group (DART) in the Computer Science Department. He is also affiliated with the Norwegian Open AILab (NAIL), hosted at NTNU. He received his PhD from the Computer Science Department at NTNU on Machine Learning for Information Retrieval. He is on the Norwegian AI Society (NAIS) board, where he organized the NAIS Symposium in 2019. He also manages the "*Machine Learning for Irregular Time Series*" research project funded by the Norwegian Research Council and coordinates other nationally funded projects in the same areas. He has more than 10 years of experience in academic and industrial research in ML and AI in different ML/AI domains. Currently, his research focuses on Deep Learning for Time Series Analysis and data efficient machine learning.

Erlend Aune (NTNU, Norway) is Associate Professor of Statistical Learning at the Norwegian University of Science and Technology (NTNU). Previous positions include Director of Data Science at fintech company Exabel and Director of Analysis and Insight at the Norwegian publisher Amedia, and CTO of HANCE, and he is the founder and organizer of Oslo Data Science Meetup. Currently, he is co-managing the ML4ITS research project funded by the

Norwegian Research Council, and he has also led several machine learning projects while working in the industry. Research interests include machine learning for time series problems, using unconventional data in time series problems, and robustness of time series models.

Claudio Gallicchio (University of Pisa, Italy) is an Associate Professor of Machine Learning at the Department of Computer Science of the University of Pisa, Italy. He received his PhD from the University of Pisa, where he focused on Recurrent and Reservoir Computing models and theory for structured data. His research is based on the fusion of concepts from Deep Learning, Recurrent Neural Networks and Randomized Neural Systems. He is the founder and former chair of the IEEE CIS Task Force on Reservoir Computing, co-founder and vice-chair of the IEEE Task Force on Randomization-based Neural Networks and Learning Systems. He is a member of the IEEE Neural Networks Technical Committee, of the IEEE CIS Task Force on Deep Learning, and of the IEEE Task Force on Learning for structured data. He (co-)authored several publications at major ML conferences, including NeurIPS, ICLR, ICML, AAAI, AISTATS, ECML-PKDD, and IJCNN/WCCI. He has (co-)organized several events (tutorials, special sessions, and workshops) at major ML international conferences. In particular, he has co-organized the Ambient AI Workshop co-located with the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2023, the International Workshop on Machine Learning for Irregular Time series (ML4ITS) co-located with the European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD) in 2021, 2023, 2024, and 2025, the workshop on Deep Learning meets Neuromorphic Hardware co-located with the ECML-PKDD 2023, 2024, and 2025, the Deep Learning in Unconventional Neuromorphic Hardware workshop within the IJCNN 2021 conference, the International Workshop on Reservoir Computing in 2019, 2024, and 2025 within the ICANN conference, and the Machine Learning and Data Mining workshop within the International Conference of the Italian Association for Artificial Intelligence from 2016 to 2021. He has a long experience in organizing special sessions at international conferences, having organized 10+ of them in recent years. He serves as a member of several program committees of conferences and workshops in ML and AI (including NeurIPS, ICML, AISTATS, ICLR, IJCAI-PRICAI, ECML-PKDD, ICONIP, IJCNN/WCCI). He is an associate editor of IEEE Transactions on Neural Networks and Learning Systems (TNNLS), and of Neural Computing and Applications (Springer). He is the coordinator of the project “NEURONE: extremely efficient NEUromorphic Reservoir cOmputing in Nanowire network hardwarE”, funded by the Italian ministry of Research, and principal investigator of the MSCA project MSCA “MINDnet: Neuromorphic computing for next-generation AI” funded by the EU.

Krzysztof Kotowski (KP Labs, Poland) is the Head of Machine Learning at KP Labs, a space-focused company and research center in Poland. With a strong background in applied computer science, he specializes in cutting-edge applications of signal processing, machine learning, and deep learning in fields that drive human progress—biomedical engineering, EEG analysis, protein folding, and space exploration. He earned his PhD from the Silesian University of Technology, Poland, where he focused on EEG signal analysis, and his MSc in Digital Signal and Image Processing from Cranfield University, UK, with a thesis in laser interferometry. His work has led to 4 patents, over 30 peer-reviewed publications, and contributions to top-ranking